

tf.compat.v1.nn.erosion2d Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/erosion2d

Computes the grayscale erosion of 4-D value and 3-D kernel tensors.

Function Signatures

```
tf.compat.v1.nn.erosion2d( value, kernel, strides, rates,  
padding, name=None )
```

Code Examples

```
tf.compat.v1.nn.erosion2d(  
value, kernel, strides, rates, padding, name=None  
)
```

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tf.compat.v1.nn.erosion2d(  
value, kernel, strides, rates, padding, name=None  
)
```

```
output[b, y, x, c] =  
min_{dy, dx} value[b,  
strides[1] * y - rates[1] * dy,  
strides[2] * x - rates[2] * dx,  
c] -  
kernel[dy, dx, c]
```

```
output[b, y, x, c] =  
min_{dy, dx} value[b,  
strides[1] * y - rates[1] * dy,  
strides[2] * x - rates[2] * dx,  
c] -  
kernel[dy, dx, c]
```

Documentation

Computes the grayscale erosion of 4-D value and 3-D kernel tensors.

The value tensor has shape [batch, in_height, in_width, depth] and the kernel tensor has shape [kernel_height, kernel_width, depth], i.e., each input channel is processed independently of the others with its own structuring function. The output tensor has shape [batch, out_height, out_width, depth]. The spatial dimensions of the output tensor depend on the padding algorithm. We currently only support the default "NHWC" data_format.

In detail, the grayscale morphological 2-D erosion is given by:

Duality: The erosion of value by the kernel is equal to the negation of the dilation of -value by the reflected kernel.

Returns

Raises